



Technical Report

Ticket Ordering with String Scores

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Jira Clone
2026-05-24

1. Executive Summary

In the Jira Clone, the order of tickets inside a sprint or the backlog used to be derived purely from each ticket's **priority** (Critical, High, Medium, Low), with the end date as a tiebreaker. That gave us broad ordering bands but **no way for a user to freely arrange tickets** — the cornerstone gesture of modern backlog tools.

This project delivers true **drag-and-drop ordering**. A single new field, `Score__c`, stores each ticket's position as a short text value. Reordering now changes **one row** regardless of how many tickets are in the list, and the order survives sprint planning sessions of any size without slowing down.

Before settling on a text-based score we evaluated the obvious alternative — a numeric position field. The study is summarised in §3 and shows why the numeric approach **cannot scale**: with whole numbers it forces a renumber on every reorder, and with decimals it runs out of precision within ~50 moves at the same spot (evidence in the supporting workbook `midpoint_sequence.xlsx`). The string-based design avoids both failure modes structurally.

2. Background / Problem Statement

2.1 What the system did before

Backlog and sprint queries returned tickets in this order:

```
ORDER BY Priority__c ASC, EndDate__c DESC
```

So tickets were grouped by priority band, with the end date as a within-band tiebreaker.

2.2 Why that was not enough

Symptom	Business cost
Users could not drag and reorder tickets	The core ceremony of backlog grooming and sprint planning was missing. Teams resorted to spreadsheets and verbal agreements.
Within a priority band, order was effectively random from the user's perspective (driven by <code>EndDate__c</code>)	"Top of the sprint" had no meaning — two tickets at the same priority would shuffle around as end dates changed.
The only way to influence position was to change priority	Priority is a <i>semantic</i> attribute, not a <i>positional</i> one. Forcing users to upgrade a ticket to "Critical" just to keep it visible at the top corrupted reporting and dashboards.
No support for stakeholders' real workflow ("put this one above that one regardless of priority")	The product felt rigid; planning meetings worked around the tool instead of through it.

2.3 What we needed

A way to express **explicit relative order** between tickets — independent of priority — that:

1. Lets the user drop a ticket anywhere in the list.
2. Survives any number of subsequent drops without degrading.
3. Costs the same to save whether the sprint has 5 tickets or 5,000.

3. Design Study — Alternatives Considered

Before committing to the chosen design, two families of approaches were studied: a **numeric position field** and a **text-based score**.

3.1 Option A — Numeric position field

The instinctive choice: give each ticket a Position__c number and sort by it. Two variants were examined.

A.1 Whole-number positions (“renumber on every move”)

If positions are integers (1, 2, 3, 4, 5), inserting a ticket between positions 1 and 2 forces the system to **shift every following ticket by one** so a slot opens up. The cost of one drag-and-drop grows with the size of the list and the platform must save many rows in a single transaction.

Property	Behaviour
Rows updated per move	N — the moved ticket and everyone after it
Time to save	Grows with sprint size
Risk of platform limit error	Real, and rising as projects grow

A.2 Decimal positions (“take the midpoint”)

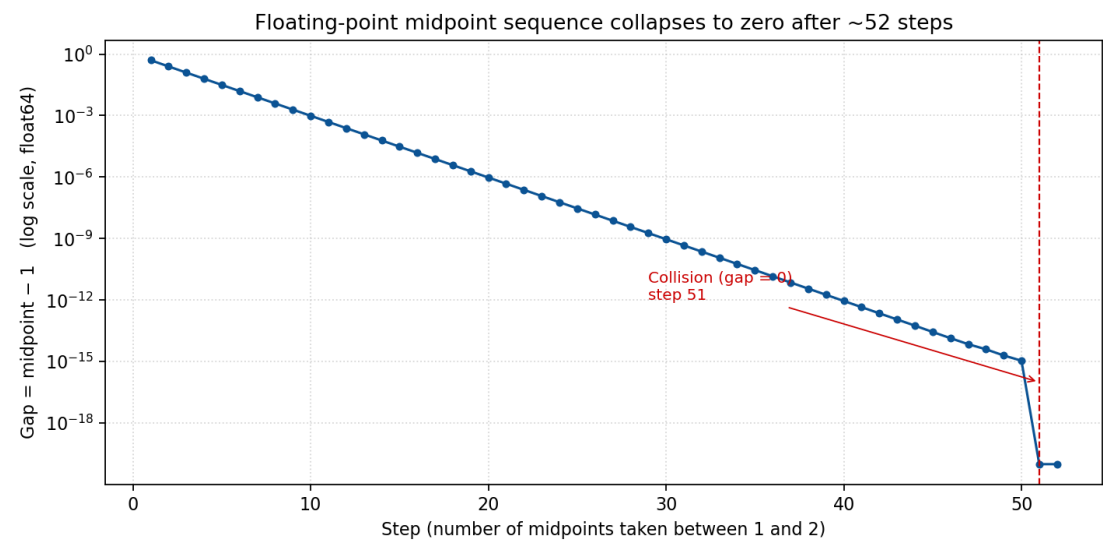
The textbook fix for integer collisions is to use decimals: drop a ticket between two others and give it the average of their positions. This buys time but does not solve the problem — **64-bit floating point precision runs out after a small number of repeated midpoints in the same neighbourhood**.

The supporting workbook `midpoint_sequence.xlsx` quantifies this. Starting from $a = 1$, $b = 2$ and repeatedly replacing b with $(a + b) / 2$, the gap between successive midpoints collapses to zero in **51 steps**:

Step	Right boundary b	Computed midpoint
1	2.0000000000000000	1.5000000000000000
10	1.0019531250000000	1.0009765625000000
25	1.000000059604645	1.000000029802322
40	1.000000000001819	1.000000000000909

Step	Right boundary b	Computed midpoint
50	1.0000000000000002	1.0000000000000001
51	1.0000000000000001	1.0000000000000000
52	1.0000000000000000	1.0000000000000000

At step 51 the midpoint collides with the lower bound — the computer can no longer represent a value strictly between them. From that point on, decimals behave like whole numbers and the system is **forced back into renumbering**.



Gap between the midpoint and the lower bound, on a logarithmic scale. The collapse to zero at step 51 is the moment decimal positions stop helping.

Roughly fifty drops in the same spot is well within what a real backlog sees over its lifetime. The decimal variant is therefore **only a deferral**, not a fix.

3.2 Option B — Text-based score (*chosen*)

A short text value, sorted alphabetically, has a property that no fixed-width number can match: **between any two text values there is always a third**. Between "0" and "1" we can use "01". Between "0" and "01" we can use "001". There is no precision limit.

Every drop becomes **one write** — only the moved ticket is updated. There is no renumbering, no collision, no precision floor.

3.3 Decision matrix

Criterion	Option A.1 (integer)	Option A.2 (decimal)	Option B — text (<i>chosen</i>)
Rows updated per move	N (whole list shifts)	1 (until precision runs out)	1

Criterion	Option A.1 (integer)	Option A.2 (decimal)	Option B — text (chosen)
Robustness over time	Degrades immediately so we need reschedule	Degrades around step 51 in any hotspot	No degradation
Risk of platform limit error	Rises with project size Eliminated	Latent, surfaces unpredictably	Eliminated
Implementation complexity	Low	Medium (precision monitoring)	Low

The text-based score wins on every axis.

4. Solution Overview

4.1 The idea in plain language

Every ticket carries a **short text label** called Score__c. The list is displayed in **alphabetical order of that label**. To place a ticket between two others, the system builds a new label that sits alphabetically between their two labels. **No other ticket has to change.**

4.2 Worked example

Start with three tickets in a sprint, created in order:

Ticket	Score
Ticket 1	0
Ticket 2	1
Ticket 3	2

Action: the user drags Ticket 3 to sit right after Ticket 1.

What the system does:

1. It looks at the ticket currently after Ticket 1 — that's Ticket 2 with score "1".
2. It joins the two scores together: "0" + "1" = "01".
3. It writes "01" to Ticket 3 only. No other ticket is touched.

Result, sorted alphabetically:

Position	Ticket	Score
1	Ticket 1	0
2	Ticket 3 (moved)	01
3	Ticket 2	1

The order is correct, the user sees the drop happen instantly, and the database touched **exactly one row**.

5. How It Works

There are only three moments in the system that involve the score.

5.1 Creating a new ticket

The ticket joins the **bottom** of its list (its sprint, or the project's backlog if it has no sprint).

Step	What happens
1	The system looks up the largest score currently in that list.
2	If the list is empty, the new score is "0".
3	Otherwise, the new score is that largest score with its last character bumped up by one (e.g. "4" → "5").

5.2 Listing tickets for the user

Step	What happens
1	The database returns tickets ordered by Score__c ascending.
2	The user sees them in that order. No client-side sorting is needed.

5.3 Reordering by drag-and-drop

The user interface sends two ticket IDs: the **moved ticket** and the **before ticket** — the ticket that should appear *just before* the moved one in the new order.

Case	What happens
The user dropped the ticket after another ticket	The system finds the ticket currently sitting just after the <i>before ticket</i> , joins those two scores, and saves that as the moved ticket's new score.
The user dropped the ticket at the top of the list (no "before")	The system takes the current first ticket's score and produces a label that sorts before it. The moved ticket gets that new label.
The <i>before ticket</i> is the last in the list	The system falls back to the "bump the last character" trick used when creating a ticket.

In every case, **only the moved ticket is updated**.

6. Business Impact / Benefits

The chosen design delivers three reinforcing benefits when measured against the numeric alternative studied in §3. Independently they matter; together they make the reorder feature **scale-proof**.

6.1 Three concrete advantages — vs. the numeric alternative

#	Advantage	Numeric (renumber)	Text score (<i>chosen</i>)
1	Database query performance	Saving a reorder updates N rows per move.	One read for the neighbour, one write for the moved ticket. Cost does not depend on sprint size.
2	In-app processing cost (CPU)	The server builds a list of every following ticket, mutates each one in memory, then dispatches the update. CPU grows with the list.	One assignment, one save. The server-side work is essentially flat.
3	Memory used per save (heap)	All shifted tickets must be held in memory at once to be saved together — heap usage grows with the sprint.	A single ticket in memory regardless of sprint size.

Advantages 2 and 3 matter **even before considering the database**, because the platform enforces strict per-transaction CPU and memory budgets. The chosen design uses a tiny, predictable slice of those budgets, leaving headroom for everything else running in the same transaction.

6.2 What this change unlocks vs. the previous Priority-based ordering

Question	Before (Priority + EndDate)	After (Score)
Can a user drag a ticket above another?	No	Yes
Is “top of the sprint” stable across edits?	No — changes to end dates re-shuffle the list	Yes — order persists until a user moves something

Question	Before (Priority + EndDate)	After (Score)
Must a user change a ticket's priority to influence its position?	Yes — corrupting priority reporting	No — position and priority are independent
Does the new feature slow down as projects grow?	n/a (feature did not exist)	No — constant cost per move

7. Risks & Considerations

Topic	What to know	Mitigation
Existing tickets need a starting score	Score__c is required, so any ticket created before this release has no value yet.	A one-time backfill is run as part of the rollout. New environments need no action.
Cross-sprint moves are a separate flow	The new function only reorders <i>within</i> a sprint or <i>within</i> a backlog. Moving a ticket from one sprint to another uses a different existing operation.	No change to existing cross-sprint move logic.
Naming of “before ticket”	The term means “ <i>the ticket that should appear before the moved one in the new order.</i> ”	Documented in the engineering appendix and in code comments.
Concatenation edge case	In a narrowly defined situation, the simple before + successor formula can produce a label that sorts in the wrong place.	The edge case is documented; a midpoint-style algorithm is the planned upgrade if it is ever observed in production.
Composite database index	The fastest query path benefits from a database index on (Sprint, Score).	The single-field index on Score__c is deployed as part of this release. The composite index will be requested through Salesforce Support if production query telemetry shows it is needed.

Technical Appendix (*For Engineers*)

The remainder of this document is for the engineering team. Non-technical readers can stop here.

A.1 Data model change

A single new field is introduced on Ticket__c:

Property	Value
API name	Score__c
Label	Score
Type	Text(255)
Required	Yes
File	force-app/main/default/objects/Ticket__c/fields/Score__c.field-meta.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
  <fullName>Score__c</fullName>
  <description>Lexicographic ordering key within a sprint or backlog.
    Assigned by Apex on insert and updated on drag/drop reorder. Sort ASC.
</description>
  <externalId>false</externalId>
  <label>Score</label>
  <length>255</length>
  <required>true</required>
  <trackTrending>false</trackTrending>
  <type>Text</type>
  <unique>false</unique>
</CustomField>
```

A.2 Previous ordering — exact SOQL

For completeness, the queries that were replaced:

```
// loadBacklogTickets – before
ORDER BY Priority__c ASC, EndDate__c DESC
```

```
// getActiveTicketsBySprintOrderByPriorityAsc – before
ORDER BY Priority__c ASC, EndDate__c DESC
```

```
// loadActiveTicketsBySprint – before
ORDER BY Priority__c ASC
```

All three now end with:

```
ORDER BY Score__c ASC NULLS LAST
```

The NULLS LAST clause is a defence in depth for any row that escaped the backfill — it will appear at the very bottom rather than confuse the order.

A.3 Indexes and lookup pattern

Three indexes work together. Only one is created explicitly:

Index	Source	Used by
Ticket__c.Id	Built-in primary key	Every controller's requireTicketExists(ticketId) call
Ticket__c.Sprint__c	Implicit — Salesforce auto-indexes every Lookup field	WHERE Sprint__c = :ticket.Sprint__c in all score helpers
Ticket__c.Score__c (single-field)	Deployable: set <externalId>true</externalId> on the field XML	ORDER BY Score__c and WHERE Score__c > :before
Sprint__c + Score__c (composite)	Not deployable — must be requested via Salesforce Support	Paged sprint listing under heavy load

Why the “look up by ID, then trust the loaded record” pattern matters

The LWC layer can send any ticket ID, but **we cannot trust the sprint ID it sends along** — client state can be stale. So no score-changing function accepts a sprintId parameter. Every flow starts by fetching the ticket by its (indexed) ID, then uses the **sprint value off the loaded record** in every subsequent query. From that point on Sprint__c is guaranteed consistent, and the indexed WHERE Sprint__c = :ticket.Sprint__c lookup is an index seek rather than a table scan.

```
public static Ticket__c requireTicketExists(String ticketId) {  
    List<Ticket__c> rows = [  
        SELECT Id, ..., Sprint__c, ..., Score__c  
        FROM Ticket__c  
        WHERE Id = :ticketId AND RecordStatus__c != 'delete'  
        LIMIT 1  
    ];  
    ...  
}
```

A.4 Where the code lives

Concern	File / symbol
Field metadata	force-app/main/default/objects/Ticket__c/fields/Score__c.field-meta.xml
Score on insert	prepareTicketForInsert + nextScoreForInsert in force-

Concern	File / symbol
	app/main/default/classes/domain/TicketService.cls
Sort by score	loadBacklogTickets, getActiveTicketsBySprintOrderByPriorityAsc, loadActiveTicketsBySprint in TicketService.cls
Reorder	moveTicketPosition(Ticket__c movedTicket, Ticket__c beforeTicket) in TicketService.cls
Helpers	incrementLastChar, previousAsciiChar, findSuccessor, findFirst, projectTicketTypeIds (all private in TicketService.cls)
Loader hydration	DomainCorrectness.requireTicketExists / requireTicketsExist updated to include Score__c
LWC-facing endpoint	ManageBacklogController.moveTicketPosition(movedTicketId, beforeTicketId)
Tests	testMoveTicketPositionWithBeforeTicket, testMoveTicketPositionWithNullBeforeTicket in TicketServiceTest.cls

A.5 The reorder algorithm

```

public static Ticket__c moveTicketPosition(Ticket__c movedTicket, Ticket__c beforeTicket) {
    // — validation —————
    if (movedTicket == null) throw new ServiceException('movedTicket is required');
    if (beforeTicket != null) {
        if (movedTicket.Id == beforeTicket.Id) {
            throw new ServiceException('movedTicket and beforeTicket must be different');
        }
        if (movedTicket.Sprint__c != beforeTicket.Sprint__c) {
            throw new ServiceException(
                'movedTicket and beforeTicket must share the same sprint (or both be in the backlog)');
        }
        if (String.isBlank(beforeTicket.Score__c)) {
            throw new ServiceException('beforeTicket has no Score__c – backfill required');
        }
    }
}

```

```

// — backlog scope (only when there is no sprint) —————
Set<Id> backlogTypeIds = null;
if (String.isBlank(movedTicket.Sprint__c)) {
    TicketType__c tt = [
        SELECT Project__c FROM TicketType__c
        WHERE Id = :movedTicket.Ticket_Type__c AND RecordStatus__c != 'deleted'
        LIMIT 1
    ];
    backlogTypeIds = projectTicketTypeIds(tt.Project__c);
}

// — compute the new score —————
String newScore;
if (beforeTicket != null) {
    Ticket__c successor = findSuccessor(movedTicket, beforeTicket.Score__c, backlogTypeIds);
    newScore = (successor == null)
        ? incrementLastChar(beforeTicket.Score__c)
        : beforeTicket.Score__c + successor.Score__c;
} else {
    Ticket__c first = findFirst(movedTicket, backlogTypeIds);
    newScore = (first == null) ? '0' : previousAsciiChar(first.Score__c);
}

// — one-row write —————
movedTicket.Score__c = newScore;
update movedTicket;
return movedTicket;
}

```

The helpers are deliberately tiny:

```

private static String incrementLastChar(String s) {
    if (String.isBlank(s)) return '0';
    Integer last = s.charAt(s.length() - 1);
    String next = String.fromCharCode(new List<Integer>{ last + 1 });
    return s.substring(0, s.length() - 1) + next;
}

private static String previousAsciiChar(String s) {
    if (String.isBlank(s)) return '0';
    Integer first = s.charAt(0);
    return String.fromCharCode(new List<Integer>{ first - 1 });
}

```

A.6 Known edge case in the concat formula

When `successor.Score__c` starts with `beforeTicket.Score__c` and continues with a character smaller than `beforeTicket.Score__c[0]`, the simple concatenation can produce a value that sorts after the successor instead of between. Example: `before = "1"`, `successor = "10"`, `concat = "110"`, but `"110" > "10"`.

This pattern does not arise from the create / move flows defined here, but if it ever does (e.g. via data migration), swap the concat for a true midpoint algorithm — the contract of `moveTicketPosition` does not need to change.

A.7 Tests delivered

Test	What it proves
<code>testMoveTicketPositionWithBeforeTicket</code>	Creating 3 tickets in a sprint yields scores "0", "1", "2"; moving ticket 3 after ticket 1 yields score "01"; the final ASC order is <code>t1 → t3 → t2</code> .
<code>testMoveTicketPositionWithNullBeforeTicket</code>	Moving ticket 3 to the top (<code>beforeTicket = null</code>) yields score "/" (ASCII 47), and ticket 3 is first under <code>ORDER BY Score__c ASC NULLS LAST</code> .

A.8 Reference

- `OBJECT_VALIDATION_LWC_APEX.md` (project root) — updated with the new field on both the LWC and apex/store rows for `Ticket__c`.
- `docs/ticket-score-ordering-notes.md` — narrative notes (this document's source).
- `midpoint_sequence.xlsx` (project root) — empirical floating-point study underlying §3.1.